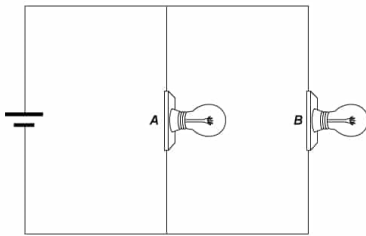


Subject:
Physics

System:
Electrostatics and Circuits

Two bulbs are connected in parallel with a battery, as shown.



Bulb A has a lower resistance than Bulb B. When the bulbs light up, which statement about the power dissipated by the bulbs is true?

- ✔ ☒ A. The power dissipated by A is greater than the power dissipated by B, because the current through bulb A is greater. [52%]
- ☐ B. The power dissipated by B is greater than the power dissipated by A, because bulb B has a greater resistance. [26%]
- ☐ C. The power dissipated by the bulbs is the same, because the bulbs have the same voltage. [11%]
- ☐ D. The power dissipated by the bulbs is the same, because bulb B has a higher resistance and a lower current than bulb A. [8%]

Correct

52% answered correctly

Explanation:

Power dissipation by lightbulbs in a parallel circuit depends on current

$P = IV$

- P Power
- I Current
- V Voltage
- R Resistance

Ohm's law

$$I = \frac{V}{R}$$
$$P = \frac{V^2}{R}$$

V across bulbs A and B is equal, but A has less resistance
Bulb A has greater current and greater power dissipation than B

© Outlines

Power P is the amount of energy used, converted, or dissipated over a time interval. In an electrical circuit, there is a change in **electrical potential energy** as charges flow across circuit elements (eg, batteries, resistors). Hence, power is equal to the product of the **current** I through the circuit element and the voltage V :

$$P = IV$$

Furthermore, **Ohm's law** implies that current flow in a resistor is equal to the ratio of V and resistance R :

$$I = \frac{V}{R}$$

Substituting into the equation for power yields:

$$P = \frac{V^2}{R}$$

In this question, bulbs A and B are in **parallel**; therefore, the voltage across each bulb is equal. Consequently, both the current through the bulbs and the power dissipated by the bulbs is **inversely proportional** to their resistance. Because bulb A has lower resistance than bulb B, bulb A has a **greater** current than bulb B. Therefore, the power dissipated by bulb A is also **greater** than in bulb B.

(Choice B) Bulb B has greater resistance than bulb A, but this implies that current flow through bulb B is less than bulb A, causing less power dissipation.

(Choice C) The voltage across the bulbs is the same; however, bulb A has less resistance. Therefore, the power dissipated by A is greater than B.

(Choice D) Bulb B does have a higher resistance than bulb A. However, less current flows through bulb B, and it dissipates less power than bulb A.

Educational objective:

The power dissipated in a circuit is the product of current and voltage. In a parallel circuit, the branch with the lowest resistance has the largest current and the largest power dissipation.

Time spent: 0 sec
Subject:
Physics

QID: 404035
System:
Electrostatics and Circuits